Twilio vs Plivo vs Telnyx

Global Virtual Numbers, IVR, SMS & WhatsApp — Complete Implementation Guide

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This guide compares three leading CPaaS (Communications Platform as a Service) providers — **Twilio**, **Plivo**, and **Telnyx** — covering global phone number availability, free trial terms, IVR (voice call) capability, SMS & WhatsApp support, number-buying cost and process, compliance/terms & conditions, and a step-by-step plan to implement any of them in a real project.

Note: Pricing changes frequently and varies heavily by destination country. Figures below are indicative (as published by third-party pricing trackers and official pages around October 2026) — always confirm the live rate on the provider's own pricing page before buying.

1. What are Twilio, Plivo and Telnyx?

All three are **CPaaS platforms** — they give you an API/SDK to programmatically send SMS, make and receive voice calls (including IVR menus), send WhatsApp messages, and rent virtual phone numbers in many countries. None of them is "voice-IVR only" — all three support Voice, SMS and WhatsApp (and more: video, fax, email in Twilio's case) on the same account.

Provider	Founded / HQ	Core Strength	Best Known For
Twilio	2008, USA	Largest global coverage, most mature docs, Studio no-code flow builder	Enterprise-grade reliability, widest country/number support
Plivo	2011, USA/India	Lower per-minute/SMS cost, strong India presence	Budget-friendly voice & SMS, good for India-US corridor
Telnyx	2009, USA	Owns its own private IP voice network (not reselling carriers)	Lowest latency voice, transparent pay-as-you-go pricing, built-in AI voice tools

2. Can you buy a Global Number? Can it do IVR?

Yes — all three let you buy virtual numbers in many countries, and every number (local, mobile, national or toll-free) can be wired to an IVR (press-1-for-sales style voice menu). Important clarification: a "global number" does not mean one number that is free to call from/to every country for free — it means:

What's True	What It Means
Receiving calls	The number can usually RECEIVE calls/SMS from callers anywhere in the world, if the caller's carrier allows it
Making calls out	To DIAL a given country, you must switch on that country's permission in your account, and you pay that country's own per-minute rate
Number type availability	Not every country offers every number type (local/toll-free/mobile) — some need local ID/address proof first

Provider	Countries with numbers	IVR on Global Numbers?	Notes
Twilio	100+ countries	Yes — via TwiML (<Gather>, <Say>, <Dial>) or drag-and-drop Studio	Widest footprint; some countries need regulatory docs (address proof, local contact, business registration)
Plivo	65+ countries	Yes — via Plivo XML (very similar to TwiML)	Self-serve plan covers only US & India by default; other countries need the Enterprise plan
Telnyx	50+ countries (US/UK/EU strong; some countries e.g. Nigeria from $25/mo)	Yes — via TeXML (accepts TwiML-style documents directly) or Call Control API	Smaller country list than Twilio but cheaper US/UK numbers; some countries need ID documents

Is a **free** global number available? **No permanent free global number exists on any of the three.** Every rented number carries a monthly rental fee. The only "free" aspect is trial credit that temporarily covers the first number + a small amount of usage during the trial period (see Section 3).

3. Free Trial — What You Actually Get

Provider	Free Credit	Card Needed?	Trial Restrictions
Twilio	Free trial bundle: ~100 SMS, ~75 voice minutes, 3,000 SendGrid emails (no fixed $ amount advertised)	No card required to start	Can call/SMS ONLY phone numbers you manually verify in the console; limited to 1 active trial number (max 3 over account lifetime); must upgrade (add card + pay) to go live for real users
Plivo	$10 free credit	No card required to start	Free-tier/self-serve usage capped at $2,500/month; geography limited to US & India only; SMS, WhatsApp and Verify API are **excluded** from free/self-serve tier — need Enterprise plan + sales contact
Telnyx	"Pretrial" gives $25 AI credit (no card); a funded/verified trial account gets $5 testing credit	No card for pretrial; card needed to unlock full sending	Mandatory business email (free Gmail/Yahoo-type addresses are rejected); KYC identity-document check required to fully verify the account before scaling up

Practical takeaway: use the free trial only to build and test your IVR/SMS flow with your own verified number. To actually serve real customers (unknown numbers, production SMS/WhatsApp, more countries) you must add a payment method and complete compliance steps — this is true for all three providers, it is industry-wide regulation, not a provider restriction.

4. IVR (Voice Menu) Capability — Voice Only, or More?

All three support full Voice IVR **and** SMS **and** WhatsApp on the same number/account (not voice-only). A typical IVR flow ("Press 1 for Sales, Press 2 for Support") is built using a simple XML-like markup returned from your webhook server:

Provider	IVR Markup Language	No-code Option	Can combine with SMS/WhatsApp?
Twilio	TwiML	Twilio Studio (drag & drop flow builder)	Yes — Programmable SMS + WhatsApp API on same account
Plivo	Plivo XML (PlivoXML)	Plivo Flow (basic visual builder)	Yes — but SMS/WhatsApp need Enterprise plan
Telnyx	TeXML (TwiML-compatible) or Call Control API (JSON webhooks, more programmatic control)	Limited; mostly code-first, plus AI Voice Assistant builder	Yes — SMS + WhatsApp Business API available

How a Simple IVR Call Flows (no code — just the logic)

Step	What Happens
1. Caller dials	Caller rings your rented number
2. Provider notifies your server	Twilio / Plivo / Telnyx sends an "incoming call" alert to your webhook URL
3. Greeting plays	Your server replies with a voice menu: "Press 1 for Sales, Press 2 for Support"
4. Caller presses a key	The key press (1 or 2) is sent back to your server
5. Call routes	Your server decides what happens next — connect to an agent, play a message, or send a follow-up SMS

5. Pricing — Number Cost, Voice, SMS, WhatsApp

5.1 Monthly Number Rental (approx., USD)

Number Type	Twilio	Plivo	Telnyx
US Local	$1.15/mo	$0.50/mo	$1.00/mo
US Toll-Free	$2.15/mo	$1.00/mo	~$1/mo (800 numbers: $500 one-time/yr then $40/mo after)
UK Local / Mobile	$3.50 / $2.50 per mo	Varies by country	Varies by country
Other int'l (e.g. Nigeria)	Varies, often $1–$5/mo where available	Varies, 65+ countries	e.g. Nigeria from $25/mo

5.2 Voice (Per-Minute, US rates as example)

Direction	Twilio	Plivo	Telnyx
Outbound call	$0.014/min	$0.010–$0.012/min	Competitive, pay-as-you-go (check live rate card)
Inbound to local number	$0.0085/min	$0.0055–$0.0085/min	Similar range
Inbound to toll-free	$0.022/min	Slightly lower than Twilio	Competitive

5.3 SMS & WhatsApp (provider markup, excludes Meta's own per-message charge for WhatsApp)

Item	Twilio	Plivo	Telnyx
SMS (US, outbound)	~$0.0079/msg	Lower than Twilio (self-serve SMS limited to US/India)	Competitive pay-as-you-go
WhatsApp provider fee	$0.005 per message (sent & received) + Meta's own rate	$0.0044 per message + Meta's rate	$0.004 per message + Meta's rate
WhatsApp calling	Supported	$0.0040/min	$0.0025/min (in & out)

*Since July 2025, Meta bills WhatsApp Business messages **per message sent** (not per 24-hour conversation window as before); from Oct 1, 2026 the first 1,000 service messages per phone number per month are free, then billed per message. The provider's fee above is **on top of** Meta's own charge.*

5.4 IVR Cost Example — 100,000 inbound + 50,000 outbound minutes/month (Twilio, US)

Item	Calculation	Cost
Inbound minutes	100,000 × $0.0085	$850
Outbound minutes	50,000 × $0.014	$700
Call recording	150,000 × $0.0025	$375
Total	Plus number rental (~$1.15/mo extra)	≈ $1,925/month

Plivo/Telnyx would typically run 20–40% lower than Twilio for the same call volume.

6. How to Buy a Number — Process, Cost & Terms and Conditions

6.1 Step-by-step buying flow (same shape on all 3 providers)

Step	What You Do
1. Sign up	Create account with email (Telnyx needs a business email, not Gmail/Yahoo-type)
2. Verify identity (KYC)	Telnyx runs a mandatory KYC document check; Twilio/Plivo ask for business details before unlocking full sending limits
3. Add payment & load balance	Add a card and top up account balance — enough to cover the number rent + expected usage
4. Search numbers	Filter by country, area code, or type (local / mobile / toll-free) in the number marketplace
5. Buy the number	Monthly rent is charged right away and auto-renews; usage (calls/SMS/WhatsApp) is billed separately as you use it
6. Complete regulatory steps	Some countries (India, Brazil, EU) need address/business proof. US SMS numbers need A2P 10DLC registration or Toll-Free Verification (free, mandatory, takes a few days)
7. Attach number to your app	Point the Voice/SMS/WhatsApp webhook URL to your server (see Section 7)

6.2 Terms & Conditions / Compliance you must accept

Requirement	Applies to	Why it exists
KYC / business verification	Telnyx (mandatory), Twilio & Plivo (for higher limits/Enterprise)	Anti-fraud / telecom regulation — prevents spam, robocalls, SIM-box fraud
A2P 10DLC registration	Any US long-code number sending application SMS	US carrier (AT&T/T-Mobile/Verizon) requirement since 2021 to curb spam texts; unregistered traffic is throttled/blocked
Toll-Free Verification	US/Canada toll-free numbers used for SMS	Free but mandatory industry-wide step before toll-free numbers can text US/Canada
Local presence documents	International numbers in India, Brazil, several EU/APAC countries	Local telecom regulators require proof the renter has a real presence/use-case in that country
Geo/Dialing Permissions	Outbound calls to certain high-risk/high-fraud destinations	Providers disable risky destinations by default to prevent toll fraud; you must explicitly enable them
Acceptable Use Policy	All three	Bans robocalls without consent, spam SMS, unsolicited WhatsApp blasts, etc. — violations can suspend the account

7. How to Implement This in Your Project — Step by Step

This same 7-step flow applies whether you choose Twilio, Plivo or Telnyx — only the SDK name and markup tags differ.

Step	What To Do
1. Create account & get keys	Sign up, verify KYC, copy your API credentials from the console. Keep them private — don't share or hard-code them.
2. Buy a number	In the console's "Buy a Number" section, search by country + capability (Voice / SMS / MMS / WhatsApp) and purchase
3. Install the SDK	Each provider has a ready-made library for Node.js, Python, PHP, Java, Ruby, Go and C# — install the one matching your language
4. Expose a webhook URL	Your server needs a public web address the provider can call when a call/SMS/WhatsApp arrives. For local testing, use a free tunnel tool (e.g. ngrok) to make your computer reachable from the internet temporarily
5. Build the IVR logic	When a call comes in, the provider contacts your webhook. Your server replies with the menu text and which key leads where (see Section 4 flow table). For multi-level menus, add one webhook per menu screen
6. Add SMS / WhatsApp (optional)	SMS: use the provider's Messages feature to send, and set up a second webhook to receive replies. WhatsApp: register a WhatsApp Business Account via Meta, get your message templates approved, then send through the same Messages feature
7. Test, then go live	Test everything on your own verified number first. Before real customers: finish the US SMS registration (A2P 10DLC / Toll-Free Verification), turn on calling permission for every country you'll dial, and keep the account balance topped up with auto-recharge

8. India-Specific: Certificates Needed & Call Pricing for IVR

8.1 Do you need a certificate/registration to make IVR calls to India?

Yes — for any commercial, automated or promotional voice call to an Indian number, TRAI's Telecom Commercial Communications Customer Preference Regulations (TCCCPR, amended 2024–2025) apply:

Requirement	What It Means
DLT registration	Your business must register as a "Principal Entity" on a DLT platform via a telecom operator, then link with a Registered Telemarketer (RTM) — usually your CPaaS provider arranges this
Number series	Service/transactional calls use 160-series (1600-series for BFSI); promotional/marketing calls use 140-series
DND/NCPR check	Promotional calls must not go to numbers on India's Do-Not-Disturb registry
Calling window	Promotional calls are generally restricted to 9 AM – 9 PM IST
Consent	Explicit recipient consent needed; violations can mean a 15-day service bar (1st offence) and 1-year blacklisting (repeat)

Separately, owning/renting an Indian local virtual number needs its own documents:

Provider	What's Needed to Get an India Number
Plivo	Dedicated India-region account + one of: GST Certificate, Certificate of Incorporation, or Udyam Registration (PAN alone not accepted). First application needs an authorized signatory's signature + company seal
Telnyx	Mobile numbers limited to 4 cities (Mumbai, Gurugram, Noida, Bangalore); needs local business/residential address proof in one of those cities + KYC
Twilio	Does NOT issue standard Indian local numbers. Twilio's own rule: outbound calls to India can only be made from international (non-Indian) numbers, and Voice OTP/2FA to India is blocked outright

*Note: the old mandatory "OSP licence" from DoT was dropped around 2020–21 and is no longer required for most cloud-telephony setups. Providers now typically operate under a **VNO (Virtual Network Operator) telecom authorisation** instead. Indian telecom rules change often — confirm the current status with DoT or a telecom-compliance lawyer before a serious production rollout.*

8.2 Call Pricing — India (per minute, approx.)

Provider	Calling an India number (local/mobile)	India number monthly rental	Notes
Twilio	Local: $0.0699/min · Mobile: $0.0496/min (official, Oct 2026)	N/A — Twilio does not issue standard Indian local numbers (see 8.1)	~4–5x higher than Twilio's US rate ($0.014/min) since Twilio has no local India carrier presence
Plivo	≈ ■0.38–■0.74/min (~$0.0046–$0.0089/min) domestic, varies by page/plan	≈ ■200/month (~$2.40)	Cheapest of the three for India because Plivo runs India-region infrastructure (India-region account required)
Telnyx	No official India-specific rate published at time of writing — general outbound SIP starts at $0.005/min + $0.002/min platform fee (US baseline)	Mobile numbers available in 4 cities (no rate confirmed here)	Check Telnyx's live per-country rate card (telnyx.com/pricing/voice-api or Mission Control Portal) before committing — India is typically priced above the US baseline

8.3 India → Other Countries (outbound international)

There is no single "international flat rate." Once a call leaves India to any other country, it is priced per that **destination country's** own per-minute rate on each platform — the same global rate table in Section 5.2 applies (e.g. calling a US number ≈ $0.014/min on Twilio, calling a UK number has its own separate rate, and so on for every country). Always check the live rate card for the exact destination before estimating cost for an India-outbound IVR campaign.

8.4 Practical note for an India-first IVR project

If your IVR's primary audience is Indian customers, pure-play global CPaaS (**Twilio**) is usually not the best fit — no local Indian number, Voice OTP blocked, and the highest per-minute India rate of the three. **Plivo** (with an India-region account) is the most India-friendly of the three covered here. **Telnyx** is expanding India coverage (mobile numbers in 4 cities as of 2026) but reportedly relies on a VNO-style authorisation — confirm scope with Telnyx support before a production rollout. For a purely India-domestic IVR/call-centre product, it is also worth comparing India-native providers such as Exotel, Ozonetel or Knowlarity (not covered in this guide), which are built specifically around DoT/TRAI compliance.

9. Package / Bundle Offers — Number + WhatsApp + SMS + MMS Together?

Short answer: none of the three sells a fixed-price "all-in-one combo" package. All three are pure pay-as-you-go — you are billed separately per number rental, per voice minute, per SMS/MMS segment, and per WhatsApp message. What differs between them is whether SMS/MMS/WhatsApp are accessible at all on the cheap self-serve tier, or gated behind a paid "Enterprise" tier.

Provider	Self-serve / pay-as-you-go tier	What needs a paid/Enterprise tier	Approx. Enterprise/Committed cost
Twilio	Voice + SMS + MMS + WhatsApp + Numbers — **all available from day 1**, no plan gate, just enable each product and pay per use	Only volume discounts / dedicated support need a sales-negotiated plan	Custom pricing via sales (no fixed published figure)
Plivo	Voice + Phone Numbers + SIP Trunking + Audio Streaming + AI Agents only — limited to US & India, capped at $2,500/month usage	**SMS, WhatsApp API, and Verify API are excluded** from self-serve — must upgrade to Enterprise to get them, plus coverage beyond US/India	≈ $1,000/month starting (third-party reports range $750–$999/mo, confirm on plivo.com/pricing)
Telnyx	Voice + SMS + MMS + Numbers + WhatsApp calling — **all available from day 1** on pay-as-you-go, similar to Twilio	Only volume discount / priority-support tiers are optional upgrades	Reported (unverified) tiers: Committed ≈ $500/mo (priority support), Growth ≈ $1,000/mo minimum, Enterprise ≈ $5,000/mo (24/7 premium support)

9.1 MMS pricing (often forgotten — picture/video messages)

Provider	Outbound MMS	Inbound MMS
Twilio	≈ $0.022/message (US) + carrier fee ≈ $0.007–$0.01	≈ $0.01–$0.02/message depending on number type
Plivo	Supported on US numbers via the Messaging API; rate not clearly published — confirm with Plivo sales	Same — confirm with Plivo sales
Telnyx	$0.015/part (US) + carrier fee	$0.005/part (US) + carrier fee

9.2 Practical Takeaway

Point	Detail
Cheapest combo	Twilio and Telnyx let you use Number + SMS + MMS + WhatsApp together on plain pay-as-you-go — no $1,000/month minimum, just turn on each product
Watch out for Plivo	Plivo's low advertised pricing covers Voice/Numbers only. The moment SMS or WhatsApp is needed, Plivo pushes you to its Enterprise plan (~$1,000/month) — which can make it the most expensive of the three overall for a combined use case
Always double-check	Plan gating and tier pricing change often — re-verify on each provider's official pricing page before committing

10. Final Comparison & Recommendation

Criteria	Twilio	Plivo	Telnyx
Global coverage	Best (100+ countries)	Good (65+, but self-serve = US/India only)	Moderate (50+), strong US/UK/EU
Pricing	Premium (highest)	Cheapest overall	Low, fully transparent pay-as-you-go
Free trial generosity	Decent bundle, no card	$10 credit, no card, but SMS/WhatsApp locked	$25 AI pretrial credit, but business email + KYC required
IVR tooling	Best (Studio no-code + TwiML)	Good (PlivoXML + basic Flow)	Good (TeXML + powerful Call Control API + AI voice tools)
SMS + WhatsApp on free/self-serve tier	Yes	No (Enterprise only)	Yes
Best for	Enterprises, widest country reach, fastest time-to-market with no-code Studio	Startups watching cost, India-heavy traffic	Developers wanting full programmatic control + lowest voice latency

Recommendation: If budget is the main constraint and most of your traffic is India/US, start with **Plivo**. If you need the widest global country coverage and a no-code IVR builder, use **Twilio**. If you want the cheapest reliable voice quality with full API control (and are comfortable building the IVR flow yourself), use **Telnyx**. All three can be swapped later since the core concepts (webhook + markup language) are nearly identical.

Sources (accessed October 2026): twilio.com/pricing, twilio.com/guidelines/in/voice, telnyx.com/pricing, plivo.com/pricing, plivo.com/voice-pricings/in, plivo.com/docs/numbers/rent-india-numbers, twilio.com/docs, developers.telnyx.com, support.plivo.com, support.twilio.com, trai.gov.in (TCCCPR regulation text), and third-party pricing/compliance trackers (cloudtalk.io, dialsheet.io, krispcall.com, smscountry.com, frejun.com). India telecom rules and CPaaS pricing both change frequently — verify current numbers and compliance status on each provider's/TRAI's own site before purchasing or launching a campaign.